Heat Diffusion Lab

FYSB21

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1 Introduction

The goal of this lab was to do a simulation of heat diffusion from a sudden temperature increase on the crust of the Earth. This was done with three tasks. In task 1, the heat diffusion in a simple one-dimensional world was simulated. The same was done in task 2a, but in a two-dimensional world. Lastly, in task 2b an thermal explosion was simulated on the crust of the earth. In this report the thermal explosion in task 2b was evaluated as the detonation of several nuclear warheads in a single spot.

The Soviet union possessed over 45000 nuclear war heads in 1986 [3]. These nuclear warheads were predominantly hydrogen bombs, which upon detonation result in a instantaneous temperature of 100 MK [5]. Accordingly, the thermal explosion was presumed to be the detonations of 45000 hydrogen bombs.

Fourier's heat equation was the central equitation of this lab [1]. This equation was solved in Fourier space, re-transformed to real space and lastly boundary conditions was applied. The solution was:

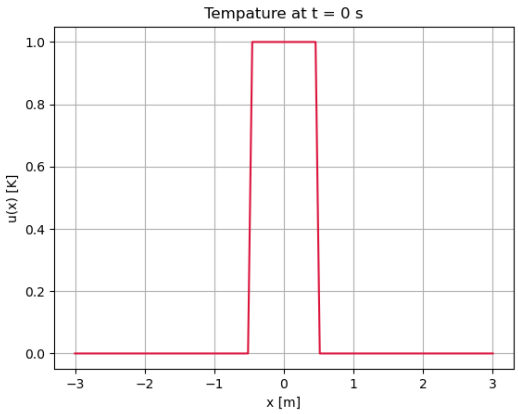
$$u(r,t) = \frac{u_0}{2}\left[\operatorname{erf}\left(\frac{r_{lim}-r}{\sqrt{4\alpha t}}\right) - \operatorname{erf}\left(\frac{-r_{lim}-r}{\sqrt{4\alpha t}}\right)\right]. \tag{1}$$

In Equation 1 u_0 is the initial temperature, r_{lim} is the radius of the initial temperature, r is the radius from the origin, α is the thermal diffusivity of the material, and t is the time passed since thermal explosion.

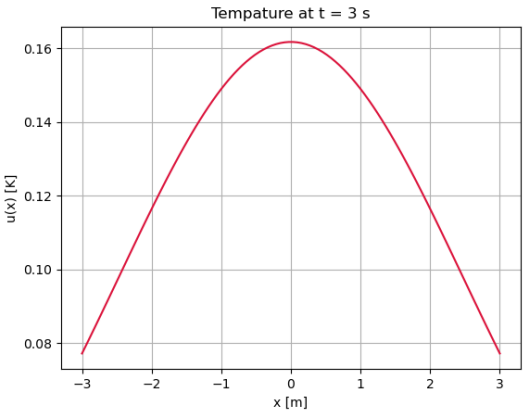
Different temperatures for the layers of the Earth had to be used. The temperature of the atmosphere, the crust, the mantel, the outer core and the inner core was used [4]. Different thermal diffusivity was also used for these different layers, were it was assumed that the thermal diffusivity increases deeper inside the earth due to a higher metal content [2],

2 Result

Two frames each for the videos regarding task 1 and task 2a can bee seen in Figure 1 and Figure 2. Two frames can also bee seen from task 2b in Figure 3. The exponential decaying function has here been chosen to have a half life of 100 million years. This is a parameter which can be changed in the simulation to a more realistic half life, but this would imply that the heat explosion would not be as visible in these frames.

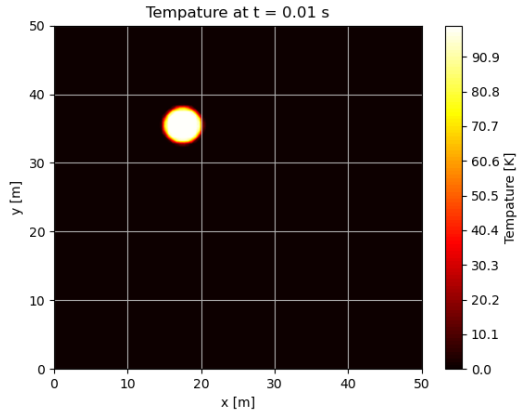


(a) The initial increase in temperature.

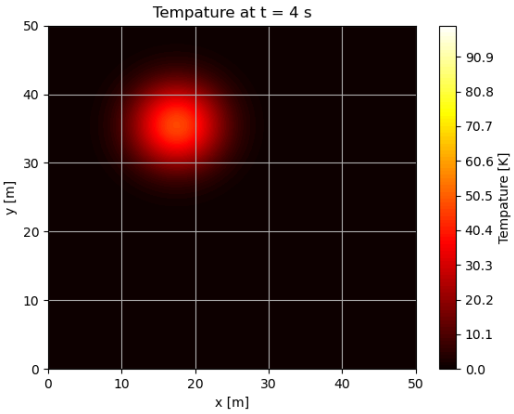


(b) The diffusion of the initial temperature after 3 s.

Figure 1: Two plots of heat diffusion in one-dimension for initial temperature of 1 K with a width of 1 m, and thermal diffusivity constant of $1\,\mathrm{m^2\,s^{-1}}$.

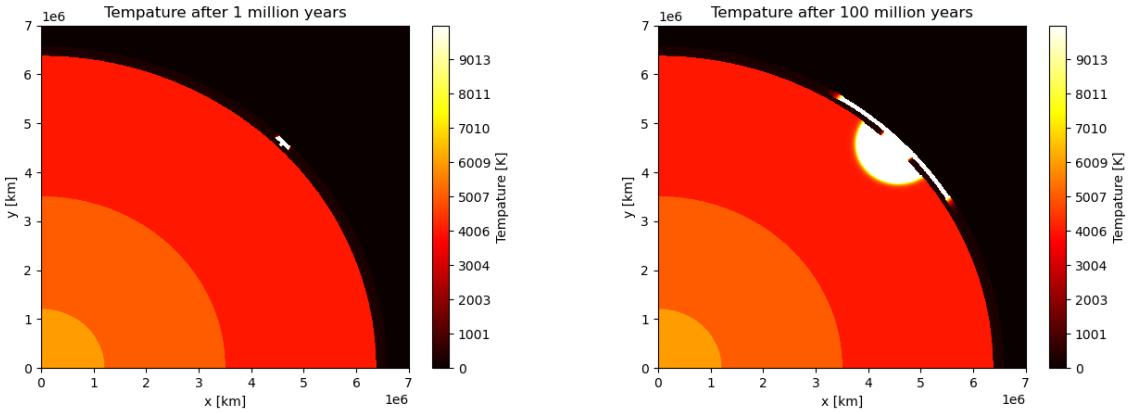


(a) The initial increase in temperature.



(b) The diffusion of the initial temperature after 4 s.

Figure 2: Two plots of heat diffusion in two-dimension with initial temperature of 100 K with a radius of 2.5 m, and a thermal diffusivity constant of $2\,\mathrm{m^2\,s^{-1}}$.



(a) The temperature soon after the detonation.

(b) The temperature after 100 million years.

Figure 3: Two plots of heat diffusion in two-dimension with initial temperature of $45000 \cdot 100\,\text{MK}$ with a radius of 10 m, and a exponential decaying function with a half life of 100 million years. The thermal diffusivity is dependent on the radius [4][2].

3 Discussion

When using a more realistic half life for the simulation, the heat dissipates faster than the speed for which it spreads. Therefor a realistic half-life imply no heat diffusion. One can therefor conclude that the heating diffused from several hydrogen bombs do not affect the Earth. Other aspects such as heat-flow, radiation, and pressure affect the earth much more.

There are several improvements which can be made to get more precise heat animations. Firstly, when solving the heat equation second order partial derivatives had to be calculated. Only the first term in Euler's approximation of the derivative was used, which is derived from the Taylor expansion of a function. Thus, when we use the second order partial derivative the error in Euler's approximation gets multiplied and therefor larger. The calculations of the second order derivative will consequently be an error.

Secondly, how the temperature and thermal diffusivity of the Earth was constructed is not very realistic. The temperature was constructed as five layers with distinct temperatures, whereas this should be a gradual change. For example, Since the thermal diffusivity gets larger closer to the core, one should predict that the heat diffusion should penetrate deeper, which it does not do in this scenario. A more precise method would be to have these variables change gradually within the Earth.

Thirdly, when simulating heat diffusion of the Earth black-body radiation, heat flow, and other aspects has to be accounted for. For example, according to black-body radiation a body will radiate more when it has a higher temperature. The radiation loss was simulated here as a exponential decaying function multiplied to Equation 1. Whilst this gives some sort of decay, it is not realistic in the way that this decay does not depend on temperature nor location inside the Earth. A better model would apply black-body radiation and thus depend on these factors.

4 Conclusion

A simulation of 45000 hydrogen bombs detonating on the earths crust was obtained through this lab. This was simulated with a solution to Fourier's heat equation and values of the thermal diffusivity of different layers of the Earth. Improvements can be made by calculating a more precise derivative, having earths layers change gradually and applying black-body radiation to the simulation. Nonetheless, it was observed that a realistic decay of the heat diffusion implies that this does not affect the Earth on a large scale.

A Appendix

A.1 Code for Task 1

```
    """
Created on Fri Dec  8 12:37:58 2023
@author: Fredrik Bergelv
"""
import numpy as np
import matplotlib.pyplot as plt
import scipy.special as scis
import matplotlib.animation as anim

#TASK 1
ylim = 1/2
alpha = 1

def h(x,t) :
    h1 = scis.erf( (ylim-x) / (np.sqrt(4*alpha*t)) )
    h2 = scis.erf( (-ylim-x) / (np.sqrt(4*alpha*t)) )
    h = 1/2*(h1 - h2)
    return h

X = np.linspace(-3, 3, 100)
t_list = 0.1*np.exp(np.linspace(0, 2*np.log(31.63), 100))-0.1
Y = h(X,t=0)
fig, ax = plt.subplots(1,1)

line, = ax.plot(X, Y, color='crimson')
plt.xlabel('x [m]')
plt.ylabel('u(x) [K]')
plt.grid()

def animate(i):
    current_h = h(X, t_list[i])
    line.set_ydata(current_h)
    plt.title(f'Tempature at t = {round(t_list[i],4)} s')

nframes = len(t_list)
animation = anim.FuncAnimation(fig, animate, frames=nframes, interval=100)

#save = anim.PillowWriter(fps=15)
#animation.save('1) video.gif', writer = save)
plt.show()
```

A.2 Code for Task 2a

```
    """
Created on Fri Dec  8 12:37:58 2023
@author: Fredrik Bergelv
"""
import numpy as np
```

```
import matplotlib.pyplot as plt
import scipy.special as scis
import matplotlib.animation as anim

#TASK 2

alpha = 2
u0=100

x0, y0, rmax = 17.5, 35.5, 2.5

def r(x,y) :
    xv  = (x - x0)**2
    yv =  (y - y0)**2
    r = np.sqrt(xv+yv)
    return r

def u(r,t) :
    h1 = scis.erf( (rmax-r) / (np.sqrt(4*alpha*t)) )
    h2 = scis.erf( (-rmax-r) / (np.sqrt(4*alpha*t)) )
    h = u0/2*(h1 - h2)
    return h

x_span = np.linspace(0, 50, 100)
y_span = np.linspace(0, 50, 100)
X,Y = np.meshgrid(x_span, y_span)
r_span = r(X, Y)
t_list = 0.1*np.exp(np.linspace(0, 2*np.log(63.25), 100))-0.1

T = u(r_span, t=0)

w = np.linspace(0, 100, 100)

fig, ax = plt.subplots(1,1)
graph = ax.contourf(X, Y, T, w, cmap='hot')
cbar = fig.colorbar(graph)
cbar.set_label('Tempature [K]')

def animate(i):
    ax.cla()
    plt.grid(True)
    plt.ylabel('y [m]')
    plt.xlabel('x [m]')
    ax.contourf(X,Y, u(r_span, t_list[i]), w,   cmap='hot')
    ax.set_title(f'Tempature at t = {round(t_list[i],4)} s')


animation = anim.FuncAnimation(fig, animate,
                               frames = len(t_list), interval=100)

plt.show()

#save = anim.PillowWriter(fps=10)
#animation.save('2b) video.gif', writer = save)

'''
plt.contourf(X, Y, T = u(r_span, t=0), cmap='jet', levels=50)
plt.xlabel('x')
```

```
plt.ylabel('y')
plt.grid(True)
plt.colorbar()
plt.show()
'''
```

A.3 Code for Task 2b

```
"""
Created on Fri Dec 8 12:37:58 2023
@author: Fredrik Bergelv
"""
import numpy as np
import matplotlib.pyplot as plt
import scipy.special as scis
import matplotlib.animation as anim

#TASK 2b

x0, y0 = (6371+98)*1000/np.sqrt(2), (6371+98)*1000/np.sqrt(2)
blast_temp, blast_radius = 45000*100*10**6, 10 #100 times 100 milion kelvin and 5x5 m^2

def r(x,y) :
    xv  = (x - x0)**2
    yv =  (y - y0)**2
    r = np.sqrt( xv + yv)
    return r

scale = 300
x_span = np.linspace(0, 7000*10**3, scale)
y_span = np.linspace(0, 7000*10**3, scale)
X,Y = np.meshgrid(x_span, y_span)

r_mesh = np.sqrt(X**2+Y**2)

radius = np.array([0,1200*10**3,3500*10**3,6371*10**3,(6371+100)*10**3,(6371+100+100)*10**3,np.inf])
alpha = np.array([0,10**(-5),5*10**(-6),10**(-6),19*10**(-6),0,0])
temp0 = np.array([6000,5000,4000,300,250,0.001,0])

alpha_E, T_E = np.meshgrid(np.zeros(scale),np.zeros(scale))

for i in range(len(radius)-1):
    Earth = np.logical_and(r_mesh >= radius[i], r_mesh < radius[i+1])
    alpha_E[Earth] = alpha[i]
    T_E[Earth] = temp0[i]


def u(r,t,alpha=alpha_E) :
    h1 = scis.erf( (blast_radius-r) / (np.sqrt(4*alpha*t)) )
    h2 = scis.erf( (-blast_radius-r) / (np.sqrt(4*alpha*t)) )
    decay = np.exp(-np.log(2)/(100*10**6*365*24*60**2)*t) #Decay of 100 million years
    h = blast_temp/2*(h1 - h2)
    return h*decay+T_E

t_list = np.linspace(0, (1800*10**6*365*24*60**2), 300) #Span of 18 billion years
```

```
levels = np.linspace(0, 10*10**3, 100)

fig, ax = plt.subplots(1,1)
graph = ax.contourf(X, Y, u(r(X,Y),0), levels, cmap='hot')
cbar = fig.colorbar(graph)
cbar.set_label('Tempature [K]')

def animate(i):
    ax.cla()
    plt.ylabel('y [km]')
    plt.xlabel('x [km]')
    ax.contourf(X,Y, u(r(X,Y), t_list[i]), levels,   cmap='hot')
    ax.set_title(f'Tempature after {round(t_list[i]/(60*60*24*365*10**6))} million years')


animation = anim.FuncAnimation(fig, animate,
                               frames = len(t_list), interval=100)

save = anim.PillowWriter(fps=30)
animation.save('2b) video.gif', writer = save)

plt.show()
```

References

[1] David Hobbs. Heat diffusion lab. *Lund University, Mathematical Physics*, 2023.

[2] Jim Wilson. Thermal diffusivity. *Electronics cooling*, 2007. https://www.electronics-cooling.com/2007/08/thermal-diffusivity/.

[3] Robert S. Norris and Hans M. Kristensen. Global nuclear stockpiles, 1945–2006. https://journals.sagepub.com/doi/pdf/10.2968/062004017#:~:text=In%201993%2C%20Victor%20Mikhailov%2C%20then,these%20warheads%20had%20been%20dismantled, *Natural Resources Defense Council,* 2006.

[4] Steven Earle, Vancover Island University. The temperature of earth's interior. https://geo.libretexts.org/Bookshelves/Geology/Physical_Geology_(Earle)/09%3A_Earths_Interior/9.02%3A_The_Temperature_of_Earths_Interior#:~:text=The%20temperature%20is%20around%201000,depending%20on%20the%20tectonic%20setting.

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